

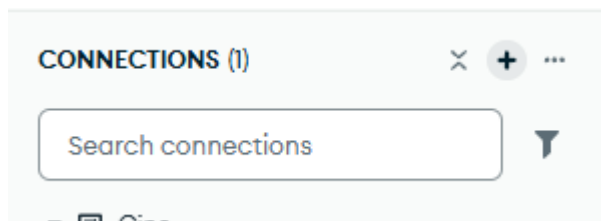
Nivell 1

Exercici 1

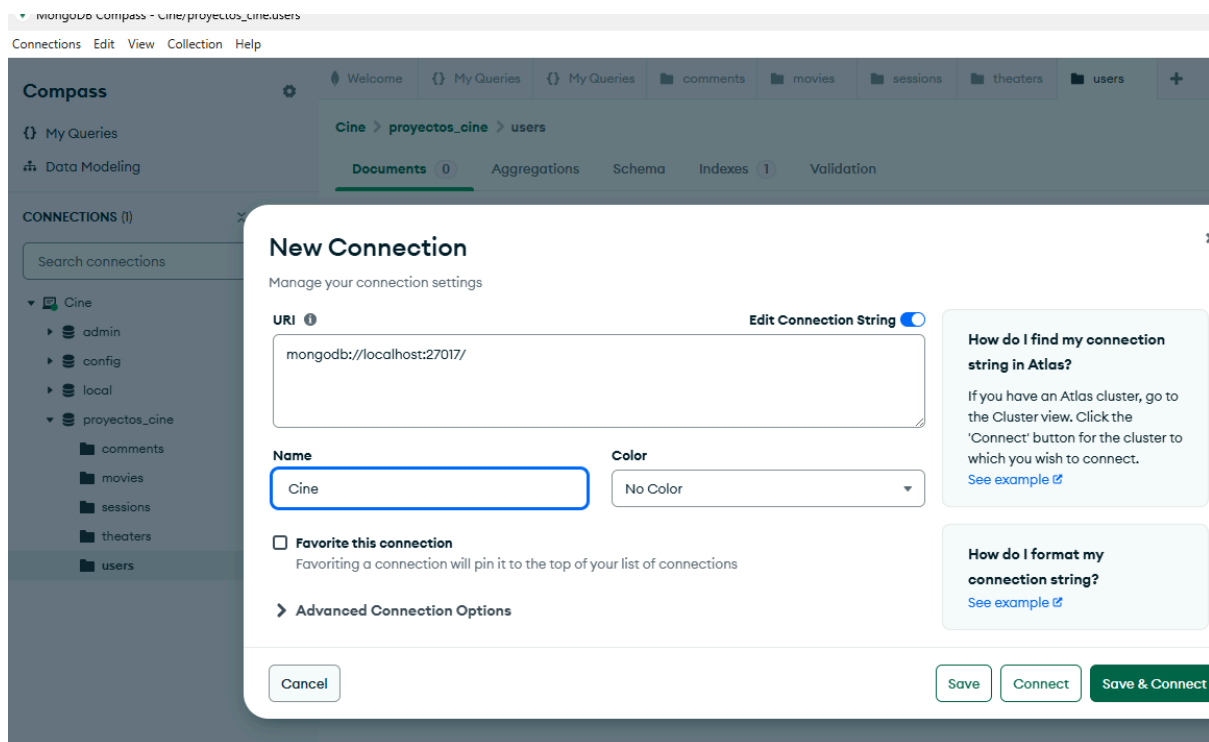
- Mostra els 2 primers comentaris que hi ha en la base de dades.
- Quants usuaris tenim registrats?
- Quants cinemes hi ha en l'estat de Califòrnia?
- Quin va ser el primer usuari/ària en registrar-se?
- Quantes pel·lícules de comèdia hi ha en la nostra base de dades?

> Creación Base de datos y Colecciones (Paso 0)

Primero le damos al botón + (tal y como se indica en la captura)



A continuación, escribimos el nombre de la nueva conexión (en este caso, 'cine')



Ahora en 'cine' le damos a su propio botón + y creamos el Database 'proyectos_cine'

Create Database

Database Name

proyectos_cine

Collection Name

☐ Time-Series

Time-series collections efficiently store sequences of measurements over a period of time. [Learn More](#)

> Additional preferences

(e.g. Custom collation, Clustered collections)

En 'proyectos_cine' le indicamos el nombre del 'Collection Name' (Por ejemplo: movies)

Create Collection

Collection Name

☐ Time-Series

Time-series collections efficiently store sequences of measurements over a period of time. [Learn More](#)

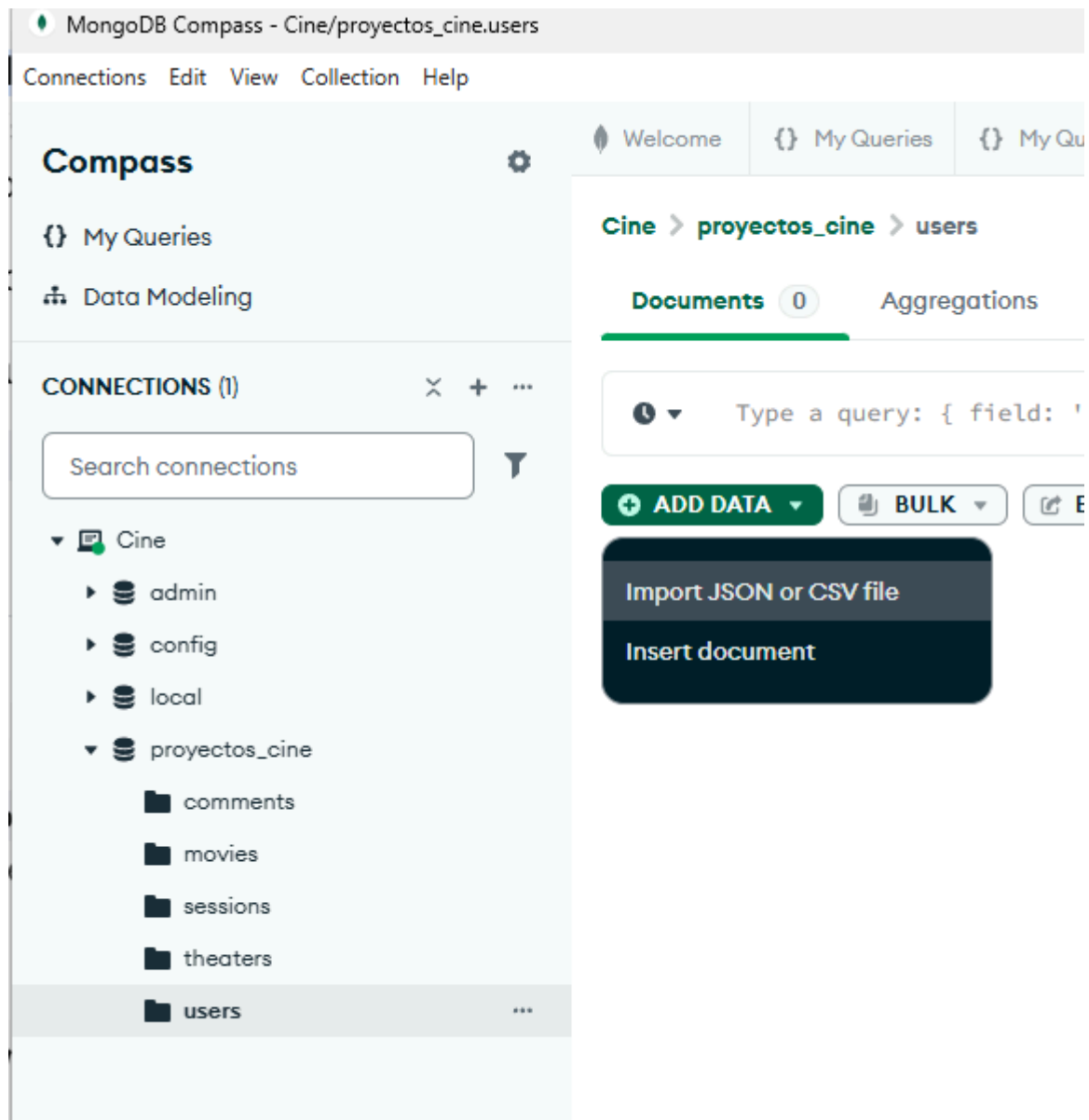
> Additional preferences

(e.g. Custom collation, Clustered collections)

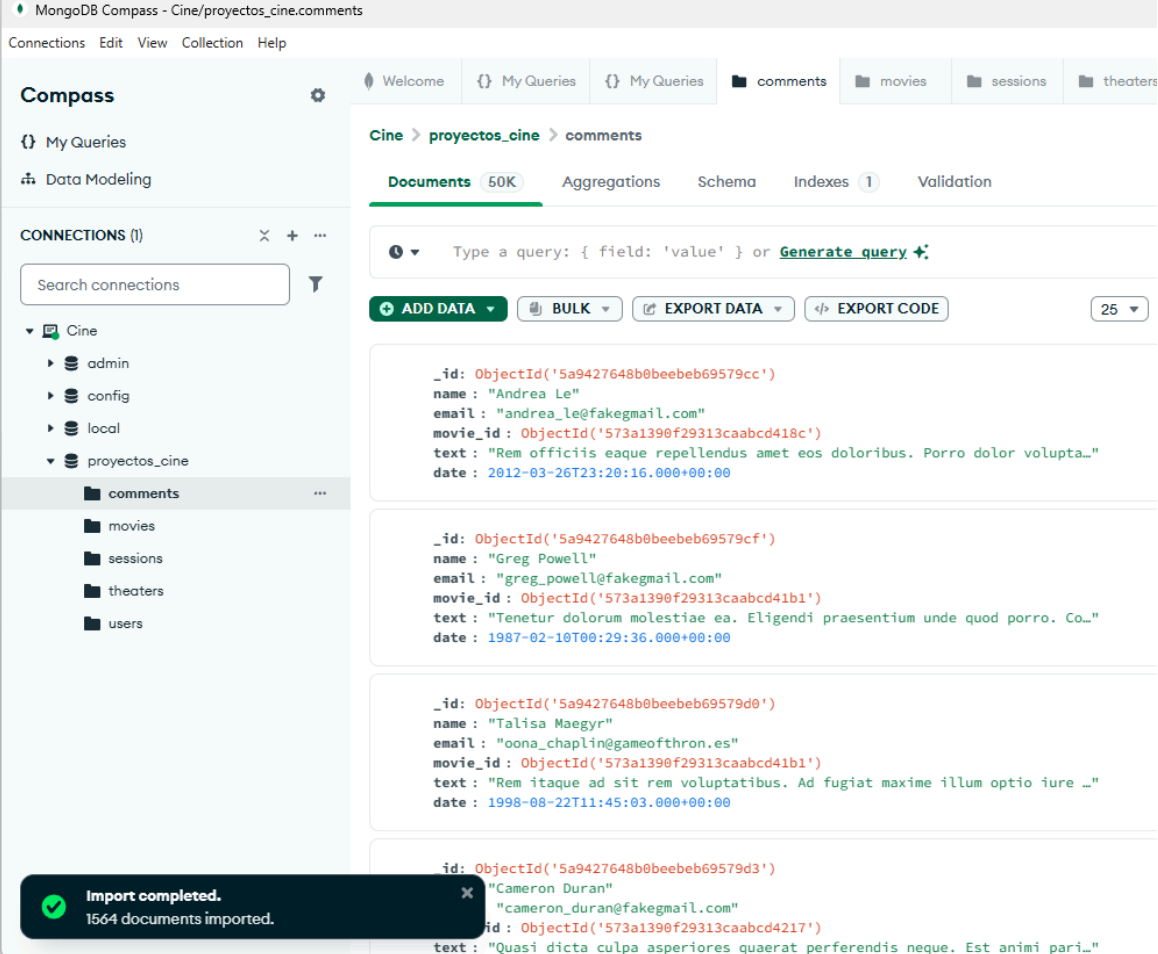
Cancel

Create Collection

Finalmente, le damos a 'ADD DATA' dentro de nuestro Collection. Importamos JSON o CSV



Y los resultados:



MongoDB Compass - Cine/proyectos_cine.comments

Connections Edit View Collection Help

Compass

My Queries

Data Modeling

CONNECTIONS (1)

Search connections

Cine

- admin
- config
- local
- proyectos_cine
 - comments
 - movies
 - sessions
 - theaters
 - users

Cine > proyectos_cine > comments

Documents 50K Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#)

ADD DATA BULK EXPORT DATA EXPORT CODE 25

```
{
  "_id": ObjectId("5a9427648b0beeb69579cc"),
  "name": "Andrea Le",
  "email": "andrea_le@fakegmail.com",
  "movie_id": ObjectId("573a1390f29313caabcd418c"),
  "text": "Rem officiis eaque repellendus amet eos doloribus. Porro dolor volupta...",
  "date": 2012-03-26T23:20:16.000+00:00
}
```

```
{
  "_id": ObjectId("5a9427648b0beeb69579cf"),
  "name": "Greg Powell",
  "email": "greg_powell@fakegmail.com",
  "movie_id": ObjectId("573a1390f29313caabcd41b1"),
  "text": "Tenetur dolorum molestiae ea. Eligendi praesentium unde quod porro. Co...",
  "date": 1987-02-10T00:29:36.000+00:00
}
```

```
{
  "_id": ObjectId("5a9427648b0beeb69579d0"),
  "name": "Talisa Maegyr",
  "email": "oona_chaplin@gameofthron.es",
  "movie_id": ObjectId("573a1390f29313caabcd41b1"),
  "text": "Rem itaque ad sit rem voluptatibus. Ad fugiat maxime illum optio iure...",
  "date": 1998-08-22T11:45:03.000+00:00
}
```

```
{
  "id": ObjectId("5a9427648b0beeb69579d3"),
  "name": "Cameron Duran",
  "email": "cameron_duran@fakegmail.com",
  "movie_id": ObjectId("573a1390f29313caabcd4217"),
  "text": "Quasi dicta culpa asperiores quaerat perferendis neque. Est animi pari..."
}
```

Import completed.
1564 documents imported.

> Mostra els 2 primers comentaris que hi ha en la base de dades.

```
_id: ObjectId('5a9427648b0beeb69579cc')
name: "Andrea Le"
email: "andrea_le@fakegmail.com"
movie_id: ObjectId('573a1390f29313caabcd418c')
text: "Rem officiis eaque repellendus amet eos doloribus. Porro dolor volupta..."
date: 2012-03-26T23:20:16.000+00:00
```

```
_id: ObjectId('5a9427648b0beeb69579cf')
name: "Greg Powell"
email: "greg_powell@fakegmail.com"
movie_id: ObjectId('573a1390f29313caabcd41b1')
text: "Tenetur dolorum molestiae ea. Eligendi praesentium unde quod porro. Co..."
date: 1987-02-10T00:29:36.000+00:00
```

> Quants usuaris tenim registrats?

Vamos al Collection de 'users' y este dato aparece de la siguiente manera:

Cine > proyectos_cine > users

Documents 185 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#) [Explain](#)

[ADD DATA](#) [BULK](#) [EXPORT DATA](#) [EXPORT CODE](#) 25 1 – 25 of 185

```
_id: ObjectId('59b99db4cfa9a34dcd7885b6')
name: "Ned Stark"
email: "sean_bean@gameofthron.es"
password: "$2b$12$UREFwsRUoyF0CRqGNK0Lz00HM/jLhgUCNNIJ9RJAqMUQ74crLJ1Vu"
```

```
_id: ObjectId('59b99db4cfa9a34dcd7885b7')
name: "Robert Baratheon"
email: "mark_addy@gameofthron.es"
password: "$2b$12$yGqxLG9LZpXA2xVDhuPnS0Zd.VURVxz7wg0LY3pn00s7u2S1Z032y"
```

```
_id: ObjectId('59b99db5cfa9a34dcd7885b8')
name: "Jaime Lannister"
email: "nikolaj_coster-waldau@gameofthron.es"
password: "$2b$12$6vz7wiw0.EI5Rilvq1zUc./9480gb1uPtXcahDxIadgyC3PS8XCUK"
```

```
_id: ObjectId('59b99db5cfa9a34dcd7885b9')
name: "Catelyn Stark"
email: "michelle_fairley@gameofthron.es"
password: "$2b$12$fiaTH5Sh1zKNFX2i/FTereWGjxoJxvmV7XL.q1fqCr8Cw0xK.mZWS"
```

Podemos verlo más al detalle en la siguiente captura:

[Explain](#)

25 1 – 25 of 185

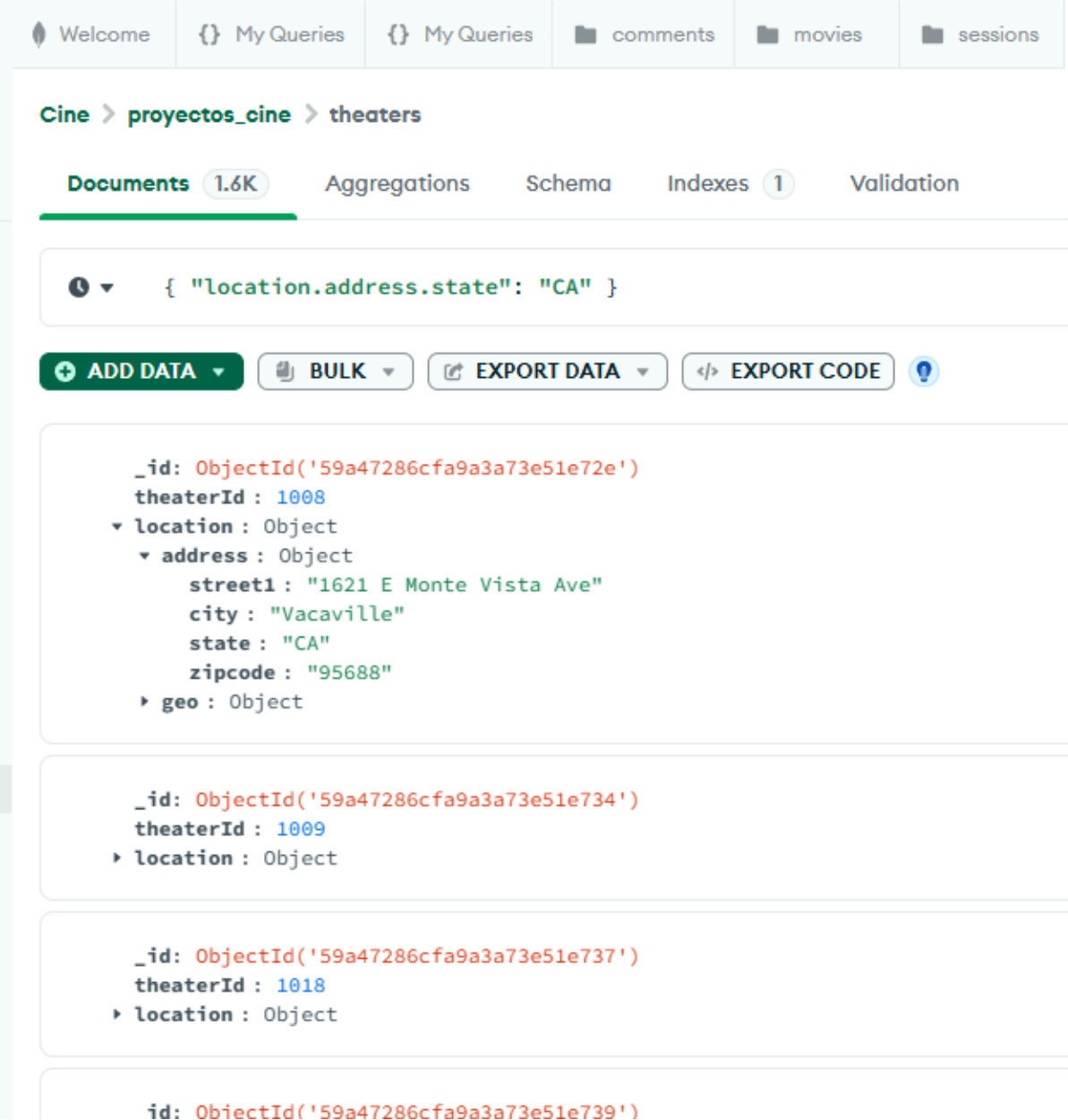
> Quants cinemes hi ha en l'estat de Califòrnia?

Ponemos el siguiente código para el JSON:

```
{ "location.address.state": "CA" }
```

Luego desplegamos el 'address' para comprobar que, efectivamente, state: "CA" (California)

5



The screenshot shows the MongoDB Atlas web interface. At the top, there's a navigation bar with tabs: Welcome, My Queries, My Queries, comments, movies, and sessions. Below this, the breadcrumb navigation shows 'Cine > proyectos_cine > theaters'. The main content area has tabs for Documents (1.6K), Aggregations, Schema, Indexes (1), and Validation. The Documents tab is selected, and a query filter is applied: { "location.address.state": "CA" }. Below the query, there are buttons for ADD DATA, BULK, EXPORT DATA, and EXPORT CODE. The query results are displayed in a list of documents. The first document is expanded, showing the 'location' field with its 'address' sub-field, which includes 'street1', 'city', 'state', and 'zipcode'. The state is confirmed to be 'CA'.

```
{ "_id": ObjectId('59a47286cfa9a3a73e51e72e'),  
  theaterId : 1008,  
  location : Object  
    address : Object  
      street1 : "1621 E Monte Vista Ave"  
      city : "Vacaville"  
      state : "CA"  
      zipcode : "95688"  
    geo : Object  
  }
```

```
{ "_id": ObjectId('59a47286cfa9a3a73e51e734'),  
  theaterId : 1009,  
  location : Object  
}
```

```
{ "_id": ObjectId('59a47286cfa9a3a73e51e737'),  
  theaterId : 1018,  
  location : Object  
}
```

```
{ id: ObjectId('59a47286cfa9a3a73e51e739')
```

> Quin va ser el primer usuari/ària en registrar-se?

El primer usuario fue Ned Stark.

Cine > proyectos_cine > users Open MongoDB shell

Documents 185 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#) ⚡

Project { field: 0 }

Sort { "_id": 1 }

Collation { locale: 'simple' }

Index Hint { field: -1 } or "indexName"

Max Time MS 60000

Skip 0 Limit 0

ADD DATA BULK EXPORT DATA EXPORT CODE

25 1 - 25 of 185

```
{
  "_id": ObjectId('59b99db4cfa9a34dcd7885b6'),
  "name": "Ned Stark",
  "email": "sean_bean@gameofthron.es",
  "password": "$2b$12$UREFwsRUoyF0CRqGNK0Lz00HM/jLhgUCNNIJ9RJAqMUQ74cr1J1Vu"
}
```

```
{
  "_id": ObjectId('59b99db4cfa9a34dcd7885b7'),
  "name": "Robert Baratheon",
  "email": "mark_addy@gameofthron.es",
  "password": "$2b$12$yGqxLG9LZpXA2xVDhuPnSOZd.VURVxz7wg0LY3pn00s7u2S1Z032y"
}
```

```
{
  "_id": ObjectId('59b99db5cfa9a34dcd7885b8'),
  "name": "Jaime Lannister",
  "email": "nikolaj_coster-waldau@gameofthron.es",
  "password": "$2b$12$6vz7wiw0.EI5Rilvq1zUc./9480gb1uPtXcahDxIadgyC3PS8XCUK"
}
```

> Quantes pel·lícules de comèdia hi ha en la nostra base de dades?

{ "genres": "Comedy" } [Generate query](#) ⚡ Explain Reset Find

ADD DATA BULK EXPORT DATA EXPORT CODE

25 1 - 25 of 7024

```
{
  "_id": ObjectId('573a1390f29313caabcd4803'),
  "plot": "Cartoon figures announce, via comic strip balloons, that they will mov...",
  "genres": Array (3)
  "runtime": 7
  "cast": Array (1)
  "num_mflix_comments": 1
  "poster": "https://m.media-amazon.com/images/M/MV5BYzg2NjNhNTctMjUxMi00ZWU4LWI3Zj_-",
  "title": "Winsor McCay, the Famous Cartoonist of the N.Y. Herald and His Moving _",
  "fullplot": "Cartoonist Winsor McCay agrees to create a large set of drawings that _",
  "languages": Array (1)
  "released": 1911-04-08T00:00:00.000+00:00
  "directors": Array (2)
  "writers": Array (2)
  "awards": Object
  "lastupdated": "2015-08-29 01:09:03.030000000"
  "year": 1911
  "imdb": Object
  "countries": Array (1)
  "type": "movie"
  "tomatoes": Object
}
```

```
{
  "_id": ObjectId('573a1390f29313caabcd50e5'),
  "plot": "The cartoonist, Winsor McCay, brings the Dinosaur back to life in th...",
  "genres": Array (3)
  "runtime": 12
  "cast": Array (3)
  "num_mflix_comments": 1
  "poster": "https://m.media-amazon.com/images/M/MV5BMT0xNzI4ODQ3NF5BMl5BanBnXkFtZT_-",
}
```

Exercici 2

Mostra'm tots els documents de les pel·lícules produïdes en 1932, però que el gènere sigui drama o estiguin en francès.

> Código de la consulta

```
{
  "year": 1932,
  "$or": [
    { "genres": "Drama" },
    { "languages": "French" }
  ]
}
```

> Resultado

Cine > proyectos_cine > movies

Documents 24K Aggregations Schema Indexes 1 Validation

```
{
  "year": 1932,
  "$or": [
    { "genres": "Drama" },
    { "languages": "French" }
  ]
}
```

ADD DATA BULK EXPORT DATA EXPORT CODE ?

```
_id: ObjectId('573a1391f29313caabcd9458')
plot: "A young artist draws a face at a canvas on his easel. Suddenly the mou..."
runtime: 55
rated: "UNRATED"
cast: Array (4)
num_mflix_comments: 1
poster: "https://m.media-amazon.com/images/M/MV5BYWY3ODE5ZWEtYjlmYi00NjA4LTk4ZW..."
title: "The Blood of a Poet"
lastupdated: "2015-09-16 13:13:05.537000000"
languages: Array (1)
released: 2010-05-20T00:00:00.000+00:00
directors: Array (1)
writers: Array (1)
awards: Object
year: 1932
imdb: Object
countries: Array (1)
type: "movie"
tomatoes: Object
```


Exercici 3

Mostra'm tots els documents de pel·lícules estatunidenques que tinguin entre 5 i 9 premis que van ser produïdes entre 2012 i 2014.

> Código

```
{
  "countries": "USA",
  "year": { "$gte": 2012, "$lte": 2014 },
  "awards.wins": { "$gte": 5, "$lte": 9 }
}
```

> Resultados

```
{
  "countries": "USA",
  "year": { "$gte": 2012, "$lte": 2014 },
  "awards.wins": { "$gte": 5, "$lte": 9 }
}
```

[+ ADD DATA](#) [BULK](#) [EXPORT DATA](#) [EXPORT CODE](#) [?](#)

```
_id: ObjectId('573a13acf29313caabd29366')
fullplot: "The manager of the negative assets sector of Life magazine, Walter Mit..."
imdb: Object
year: 2013
plot: "When his job along with that of his co-worker are threatened, Walter t..."
genres: Array (3)
rated: "PG"
metacritic: 54
title: "The Secret Life of Walter Mitty"
lastupdated: "2015-08-31 00:10:51.747000000"
languages: Array (3)
writers: Array (3)
type: "movie"
tomatoes: Object
poster: "https://m.media-amazon.com/images/M/MV5BODYwNDYxNDk1Nl5BMl5BanBnXkFtZT..."
num_mflix_comments: 1
released: 2013-12-25T00:00:00.000+00:00
awards: Object
countries: Array (2)
cast: Array (4)
directors: Array (1)
runtime: 114
```

```
_id: ObjectId('573a13b5f29313caabd45772')
plot: "After their cave is destroyed, a caveman family must trek through an u..."
genres: Array (3)
runtime: 98
metacritic: 55
rated: "PG"
cast: Array (4)
poster: "https://m.media-amazon.com/images/M/MV5BMTcyOTc2OTA1Ml5BMl5BanBnXkFtZT..."
title: "The Croods"
fullplot: "The Croods is a prehistoric comedy adventure that follows the world's ..."
languages: Array (1)
released: 2013-03-22T00:00:00.000+00:00
directors: Array (2)
writers: Array (5)
awards: Object
lastupdated: "2015-09-10 17:45:48.560000000"
year: 2013
imdb: Object
countries: Array (1)
type: "movie"
tomatoes: Object
```

Nivell 2

Exercici 1

Compte quants comentaris escriu un usuari/ària que utilitza "GAMEOFTHRON.ES" com a domini de correu electrònic.

> Código de la consulta

```
{
  "email": { "$regex": "@GAMEOFTHRON\\.ES$", "$options": "i" }
}
```

> Resultado

```
{
  "email": { "$regex": "@GAMEOFTHRON\\.ES$", "$options": "i" }
}
```

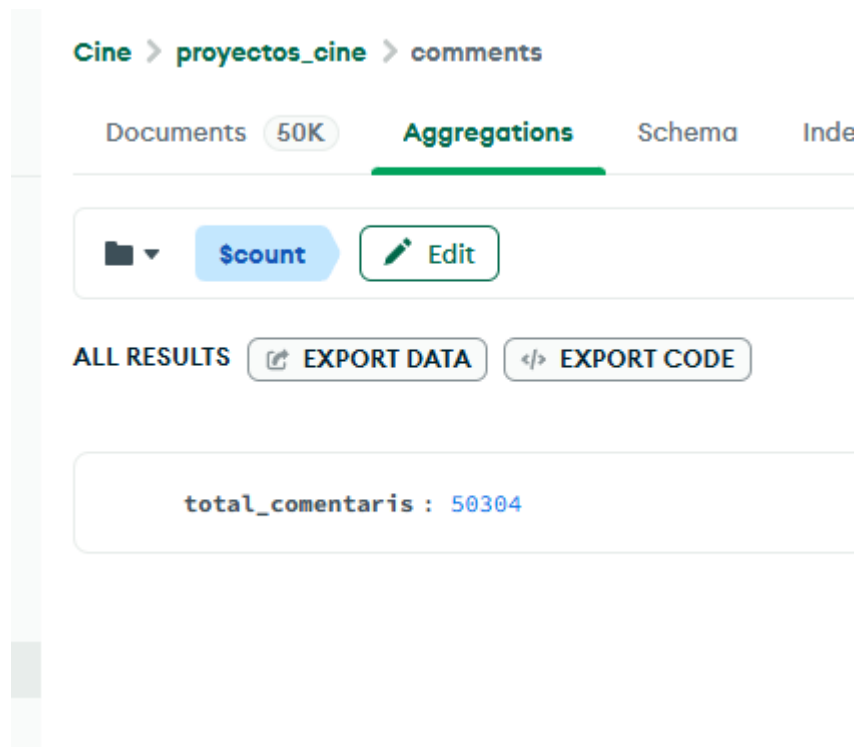
[+ ADD DATA ▾](#)[BULK ▾](#)[EXPORT DATA ▾](#)[EXPORT CODE](#)

```
_id: ObjectId('5a9427648b0beebe69579d0')
name: "Talisa Maegyr"
email: "oona_chaplin@gameofthron.es"
movie_id: ObjectId('573a1390f29313caabcd41b1')
text: "Rem itaque ad sit rem voluptatibus. Ad fugiat maxime illum optio iure ..."
date: 1998-08-22T11:45:03.000+00:00
```

```
_id: ObjectId('5a9427648b0beebe69579d5')
name: "Petyr Baelish"
email: "aidan_gillen@gameofthron.es"
movie_id: ObjectId('573a1390f29313caabcd4218')
text: "Quo deserunt ipsam ipsum. Tenetur eos nemo nam sint praesentium minus ..."
date: 2001-07-13T19:25:09.000+00:00
```

```
_id: ObjectId('5a9427648b0beebe69579db')
name: "Olly"
email: "brenock_o'connor@gameofthron.es"
movie_id: ObjectId('573a1390f29313caabcd413b')
text: "Perspiciatis sit nariatur quas. Perferendis officia harum insum delenit..."
```

Y luego añadimos un Stage con COUNT “comentarios”



En este caso, MongoDB no necesita que le digamos qué contar porque, al estar utilizando un sistema de Aggregation el count cuenta automáticamente todo lo que le llega del paso anterior.

Exercici 2

Quants cinemes hi ha en cada codi postal situats dins de l'estat Washington D. C. (DC)?

> Filtros con Aggregation (Previsualización)

Como es una consulta un poco más compleja, esta vez lo haremos desde aggregation.

Stage 1: { "location.address.state": "DC" }

Stage 2: {
 "_id": "\$location.address.zipcode",
 "total_cinemes": { "\$sum": 1 }
}

Cine > proyectos_cine > theaters

Documents 1.6K Aggregations Schema Indexes 1 Validation

▮ ▾ \$match \$group

Collation {locale: 'simple'}

Untitled - modified [SAVE](#) [+ CREATE NEW](#) [EXPORT DATA](#) [EXPORT CODE](#)

▼ 1544 Documents in the collection

Preview of documents

```
{
  "_id": ObjectId("59a47286cfa9a3a73e51e72c"),
  "theaterId": 1000,
  "location": Object
}
```

```
{
  "_id": ObjectId("59a47286cfa9a3a73e51e72d"),
  "theaterId": 1003,
  "location": Object
}
```

```
{
  "_id": ObjectId("59a47286cfa9a3a73e51e72e"),
  "theaterId": 1008,
  "location": Object
}
```

```
{
  "_id": ObjectId("59a47286cfa9a3a73e51e72f"),
  "theaterId": 1004,
  "location": Object
}
```

Stage 1 \$match ☒

```
1 { "location.address.state": "DC" }
```

Output preview after \$match stage (Sample of 3 documents)

```
{
  "_id": ObjectId("59a47286cfa9a3a73e51e785"),
  "theaterId": 1092,
  "location": Object
}
```

```
{
  "_id": ObjectId("59a47287cfa9a3a73e51ec33"),
  "theaterId": 801,
  "location": Object
}
```

```
{
  "_id": ObjectId("59a47287cfa9a3a73e51ecd8"),
  "theaterId": 8500,
  "location": Object
}
```

Stage 2 \$group ☒

```
1 {
2   "_id": "$location.address.zipcode",
3   "total_cinemes": { "$sum": 1 }
4 }
```

Output preview after \$group stage (Sample of 3 documents)

```
{
  "_id": "20016",
  "total_cinemes": 1
}
```

```
{
  "_id": "20010",
  "total_cinemes": 1
}
```

```
{
  "_id": "20002",
  "total_cinemes": 1
}
```

> Resultado

Clicamos en RUN y veremos como ha salido justo como esperábamos.

Cine > proyectos_cine > theaters

Documents 1.6K Aggregations Schema Indexes 1 Validation

▮ ▾ \$match \$group [Edit](#)

ALL RESULTS [EXPORT DATA](#) [EXPORT CODE](#)

```
{
  "_id": "20002",
  "total_cinemes": 1
}
```

```
{
  "_id": "20010",
  "total_cinemes": 1
}
```

```
{
  "_id": "20016",
  "total_cinemes": 1
}
```

Lo que hace es coger cada vez que ha tenido que agrupar un cine, y sumar 1. Como solo hay 1 cine en cada Código Postal, aparece 1 en los tres casos.

Nivell 3

Exercici 1

Troba totes les pel·lícules dirigides per John Landis amb una puntuació IMDb (Internet Movie Database) d'entre 7,5 i 8.

> Código de la consulta

```
{
  "directors": "John Landis",
  "imdb.rating": { "$gte": 7.5, "$lte": 8 }
}
```

> Resultado

```
{
  "directors": "John Landis",
  "imdb.rating": { "$gte": 7.5, "$lte": 8 }
}
```

[+ ADD DATA](#) [BULK](#) [EXPORT DATA](#) [EXPORT CODE](#) [?](#)

```
_id: ObjectId('573a1397f29313caabce6d94')
fullplot: "Faber College has one frat house so disreputable it will take anyone. ..."
imdb: Object
year: 1978
plot: "At a 1962 college, Dean Vernon Wormer is determined to expel the entire..."
genres: Array (1)
rated: "R"
metacritic: 82
title: "Animal House"
lastupdated: "2015-09-13 00:02:47.803000000"
languages: Array (2)
writers: Array (3)
type: "movie"
tomatoes: Object
poster: "https://m.media-amazon.com/images/M/MV5BM2M2ZDA4MTYtOGRjMi00OTg5LWI1ZT..."
num_mflix_comments: 1
released: 1978-07-28T00:00:00.000+00:00
awards: Object
countries: Array (1)
cast: Array (4)
directors: Array (1)
runtime: 109
```

```
▶ _id: ObjectId('573a1397f29313caabce76f7')
plot: "Jake Blues, just out from prison, puts together his old band to save t..."
genres: Array (3)
runtime: 133
rated: "R"
cast: Array (4)
num_mflix_comments: 1
poster: "https://m.media-amazon.com/images/M/MV5BYTdldmExOGUtN2I3MS00MjY5LWE1NT..."
title: "The Blues Brothers"
fullplot: "After the release of Jake Blues from prison, he and brother Elwood go ..."
languages: Array (1)
released: 1980-06-20T00:00:00.000+00:00
directors: Array (1)
writers: Array (2)
awards: Object
lastupdated: "2015-09-01 00:09:39.130000000"
year: 1980
imdb: Object
countries: Array (1)
type: "movie"
tomatoes: Object
```

Exercici 2

Mostra en un mapa la ubicació de tots els teatres de la base de dades.

> Resultado

Type a query: { field: 'value' } or [Generate query](#)

EXPORT SCHEMA

address

document

Document with 5 nested fields.

geo

coordinates

